



Lab 03

Configure Cisco Routers for Syslog, NTP, and SSH Operations

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3.1 Objective

The Objectives of this lab are:

- Configure OSPF MD5 authentication.
- Configure NTP.
- Configure routers to log messages to the syslog server.
- Configure R3 to support SSH connections.

3.2 Introduction

In this lab, you will configure OSPF MD5 authentication for secure routing updates

The NTP Server is the master NTP server in this lab. You will configure authentication on the NTP server and the routers. You will configure the routers to allow the software clock to be synchronized by NTP to the time server. Also, you will configure the routers to periodically update the hardware clock with the time learned from NTP.

The Syslog Server will provide message logging in this lab

. You will configure the routers to identify the remote host (Syslog server) that will receive logging messages.

You will need to configure timestamp service for logging on the routers. Displaying the correct time and date in Syslog messages is vital when using Syslog to monitor a network.

You will configure R3 to be managed securely using SSH instead of Telnet. The servers have been pre-configured for NTP and Syslog services respectively. NTP will not require authentication. The routers have been pre-configured with the following passwords:

Enable password: **ciscoenpa55**

Password for vty lines: **ciscovtypa55**

Note: Note: MD5 is the strongest encryption supported in the version of Packet Tracer used to develop this lab. Although MD5 has known vulnerabilities, you should use the encryption that meets the security requirements of your organization. In this lab, the security requirement specifies MD5.

Topology

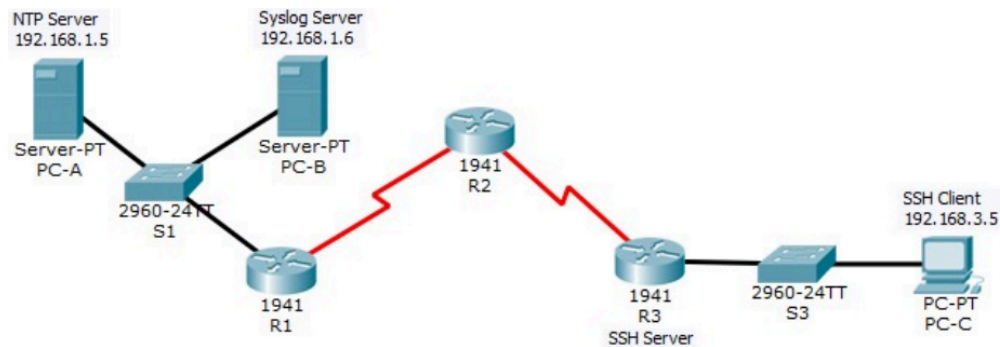


Figure 1: Topology

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.5	255.255.255.0	192.168.1.1	S1 F0/6
PC-B	NIC	192.168.1.6	255.255.255.0	192.168.1.1	S2 F0/18
PC-C	NIC	192.168.3.5	255.255.255.0	192.168.3.1	S3 F0/18

Table 1: Addressing Table

Task 1: Configure OSPF MD5 Authentication

1. Construct the topology as shown in Figure 1. Provide addresses to all devices using the values given in Table 1. Test connectivity. All devices should be able to ping all other IP addresses.
2. Configure OSPF MD5 authentication for all the routers in area 0.

```
R1(config)# router ospf 1
R1(config-router)# area 0 authentication message-digest
```



```
GigabitEthernet0/1 is up, line protocol is up
  Internet address is 192.168.1.1/24, Area 0
  Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.1.1, Interface address 192.168.1.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    No key configured, using default key id 0
Serial0/0/0 is up, line protocol is up
  Internet address is 10.1.1.1/30, Area 0
  Process ID 1, Router ID 192.168.1.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:03
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 10.2.2.2
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    Youngest key id is 1
"
```



```
R2#show ip ospf interface
```

```
Serial0/0/1 is up, line protocol is up
  Internet address is 10.2.2.2/30, Area 0
  Process ID 1, Router ID 10.2.2.2, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:06
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 192.168.3.1
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    No key configured, using default key id 0
Serial0/0/0 is up, line protocol is up
  Internet address is 10.1.1.2/30, Area 0
  Process ID 1, Router ID 10.2.2.2, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:04
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 192.168.1.1
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    Youngest key id is 1
```



```
R3#show ip ospf interface
```

```
GigabitEthernet0/1 is up, line protocol is up
  Internet address is 192.168.3.1/24, Area 0
  Process ID 1, Router ID 192.168.3.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.3.1, Interface address 192.168.3.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
  No key configured, using default key id 0
Serial10/0/1 is up, line protocol is up
  Internet address is 10.2.2.1/30, Area 0
  Process ID 1, Router ID 192.168.3.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  Hello due in 00:00:01
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
  Adjacent with neighbor 10.2.2.2
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
  No key configured, using default key id 0
R3#
```

3. Configure the MD5 key for all the routers in area 0. Use the password **MD5pa55** for key

```
R1(config)# interface s0/0/0
R1(config-if)# ip ospf message-digest-key 1 md5 MD5pa55
```



```
R1#show ip ospf interface

GigabitEthernet0/1 is up, line protocol is up
 Internet address is 192.168.1.1/24, Area 0
 Process ID 1, Router ID 192.168.1.1, Network Type BROADCAST, Cost: 1
 Transmit Delay is 1 sec, State DR, Priority 1
 Designated Router (ID) 192.168.1.1, Interface address 192.168.1.1
 No backup designated router on this network
 Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
   No Hellos (Passive interface)
 Index 1/1, flood queue length 0
 Next 0x0(0)/0x0(0)
 Last flood scan length is 1, maximum is 1
 Last flood scan time is 0 msec, maximum is 0 msec
 Neighbor Count is 0, Adjacent neighbor count is 0
 Suppress hello for 0 neighbor(s)
 Message digest authentication enabled
   No key configured, using default key id 0
Serial0/0/0 is up, line protocol is up
 Internet address is 10.1.1.1/30, Area 0
 Process ID 1, Router ID 192.168.1.1, Network Type POINT-TO-POINT, Cost: 64
 Transmit Delay is 1 sec, State POINT-TO-POINT,
 Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
   Hello due in 00:00:08
 Index 2/2, flood queue length 0
 Next 0x0(0)/0x0(0)
 Last flood scan length is 1, maximum is 1
 Last flood scan time is 0 msec, maximum is 0 msec
 Neighbor Count is 1 , Adjacent neighbor count is 1
   Adjacent with neighbor 10.2.2.2
 Suppress hello for 0 neighbor(s)
 Message digest authentication enabled
   Youngest key id is 1
R1#
```



Password:

R2#show ip ospf interface

```
Serial0/0/1 is up, line protocol is up
  Internet address is 10.2.2.2/30, Area 0
  Process ID 1, Router ID 10.2.2.2, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:01
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 192.168.3.1
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    No key configured, using default key id 0
Serial0/0/0 is up, line protocol is up
  Internet address is 10.1.1.2/30, Area 0
  Process ID 1, Router ID 10.2.2.2, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:09
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 192.168.1.1
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    Youngest key id is 1
```



```
R3>en
Password:
R3#show ip ospf interface

GigabitEthernet0/1 is up, line protocol is up
  Internet address is 192.168.3.1/24, Area 0
  Process ID 1, Router ID 192.168.3.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 192.168.3.1, Interface address 192.168.3.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    No key configured, using default key id 0
Serial10/0/1 is up, line protocol is up
  Internet address is 10.2.2.1/30, Area 0
  Process ID 1, Router ID 192.168.3.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:08
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 10.2.2.2
  Suppress hello for 0 neighbor(s)
  Message digest authentication enabled
    No key configured, using default key id 0
R3#
```

4. Verify configurations.
 - a. Verify the MD5 authentication configurations using the commands **show ip ospf interface**.
 - b. Verify end-to-end connectivity.

Task 2: Configure NTP

- 1) Enable NTP authentication on PC-A.
 - a. On **PC-A**, click **NTP** under the Services tab to verify NTP service is enabled.
 - b. To configure NTP authentication, click **Enable** under Authentication. Use key **1** and password **NTPpa55** for authentication.
- 2) Configure R1, R2, and R3 as NTP clients. Verify client configuration using the command **show ntp status**. **Hint:** Use the **ntp server** command.



```
R1#show ntp status
Clock is synchronized, stratum 2, reference is 192.168.1.5
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is ECFE8337.0000032C (16:58:31.812 UTC Fri Jan 30 2026)
clock offset is 0.00 msec, root delay is 0.00 msec
root dispersion is 10.20 msec, peer dispersion is 0.24 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system poll
interval is 5, last update was 6 sec ago.
R1#

R2#show clock
17:0:59.17 UTC Fri Jan 30 2026
R2#
R2#show ntp status
Clock is synchronized, stratum 2, reference is 192.168.1.5
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is ECFE8429.00000310 (17:2:33.784 UTC Fri Jan 30 2026)
clock offset is 0.00 msec, root delay is 5.00 msec
root dispersion is 10.92 msec, peer dispersion is 0.12 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system poll
interval is 4, last update was 11 sec ago.
R2#

R3#show ntp status
Clock is synchronized, stratum 2, reference is 192.168.1.5
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is ECFE84CF.000001EB (17:5:19.491 UTC Fri Jan 30 2026)
clock offset is 14.00 msec, root delay is 33.00 msec
root dispersion is 10.51 msec, peer dispersion is 0.12 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system poll
interval is 4, last update was 9 sec ago.
R3#
```

- 3) Configure routers to update hardware clock. **Hint:** use the **ntp update-calendar** command.
- 4) Configure NTP authentication on **R1**, **R2**, and **R3** using key **1** and password **NTPpa55**.

- o R1(config)# **ntp authenticate**
- o R1(config)# **ntp trusted-key 1**
- o R1(config)# **ntp authentication-key 1 md5 NTPpa55**

```
R1(config)#ntp authenticate
R1(config)#ntp authentication-
R1(config)#ntp authentication-key 1 md5 NTPpa55
R1(config)#ntp trusted-key 1
```

```
R2(config)#ntp authenticate
R2(config)#ntp authentication-key 1 md5 NTPpa55
R2(config)#ntp trusted-key 1
```

```
R3(config)#ntp authenticate
R3(config)#ntp authentication-key 1 md5 NTPpa55
R3(config)#ntp trusted-key 1
```

- 5) Configure routers to timestamp log messages.

```
R1(config)# service timestamps log datetime msec
```



```
R1(config)#service timestamps log datetime msec
```

```
R2(config)#service timestamp log datetime msec
```

```
R3(config)#service timestamp log datetime msec
```

Task 3: Configure Routers to Log Messages to the Syslog Server

1. Configure the routers to identify the remote host (Syslog Server) that will receive logging messages. On PC-B, click Syslog under the Services tab to verify syslog service is enabled.

Hint: use the **logging host** command.

The router console will display a message that logging has started.

```
R1#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
                  0 flushes, 0 overruns, xml disabled, filtering disabled)
```

```
No Active Message Discriminator.
```

```
No Inactive Message Discriminator.
```

```
Console logging: level debugging, 19 messages logged, xml disabled,
                  filtering disabled
Monitor logging: level debugging, 19 messages logged, xml disabled,
                  filtering disabled
Buffer logging:  level debugging, 0 messages logged, xml disabled,
                  filtering disabled
```

```
Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled
```

```
No active filter modules.
```

```
ESM: 0 messages dropped
Trap logging: level informational, 19 message lines logged
Logging to 192.168.1.6 (udp port 514, audit disabled,
                    authentication disabled, encryption disabled, link up),
                    2 message lines logged,
                    0 message lines rate-limited,
                    0 message lines dropped-by-MD,
                    xml disabled, sequence number disabled
                    filtering disabled
```

```
Log Buffer (4096 bytes):
```



```
R2(config)#logging host 192.168.1.6
R2(config)#exit
R2#
*Jan 30, 17:15:33.1515: SYS-5-CONFIG_I: Configured from console by console
*Jan 30, 17:15:33.1515: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.6 p
- CLI initiated
R2#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
                  0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 19 messages logged, xml disabled,
                  filtering disabled
Monitor logging: level debugging, 19 messages logged, xml disabled,
                  filtering disabled
Buffer logging:  level debugging, 0 messages logged, xml disabled,
                  filtering disabled

Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped
Trap logging: level informational, 19 message lines logged
Logging to 192.168.1.6 (udp port 514, audit disabled,
authentication disabled, encryption disabled, link up),
2 message lines logged,
0 message lines rate-limited,
0 message lines dropped-by-MD,
xml disabled, sequence number disabled
filtering disabled

Log Buffer (4096 bytes):
```



```
R3#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
                  0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 11 messages logged, xml disabled,
                  filtering disabled
Monitor logging: level debugging, 11 messages logged, xml disabled,
                  filtering disabled
Buffer logging:  level debugging, 0 messages logged, xml disabled,
                  filtering disabled

Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

--More--
```

2. Verify logging configuration. Use the command **show logging** to verify logging has been enabled
3. Examine logs of the Syslog Server. From the **Services** tab of the **Syslog Server**'s dialogue box, select the **Syslog services** button. Observe the logging messages received from the routers. **Note:** Log messages can be generated on the server by executing commands on the router. For example, entering and exiting global configuration mode will generate an informational configuration message. You may need to click a different service and then click **Syslog** again to refresh the message display.

Syslog			
Syslog			
Service			☒ On ☐ Off
	Time	HostName	Message
1	01.30.2026 05:10:42.320 PM	192.168.1.1	%SYS-5-CONFIG_I: Configured from console by console
2	01.30.2026 05:10:42.320 PM	192.168.1.1	: %SYS-6-LOGGINGHOST_STARTSTOP...
3	01.30.2026 05:15:33.920 PM	10.1.1.2	%SYS-5-CONFIG_I: Configured from console by console
4	01.30.2026 05:15:33.920 PM	10.1.1.2	: %SYS-6-LOGGINGHOST_STARTSTOP...
5	01.30.2026 05:17:40.097 PM	10.2.2.1	%SYS-5-CONFIG_I: Configured from console by console
6	01.30.2026 05:17:40.097 PM	10.2.2.1	: %SYS-6-LOGGINGHOST_STARTSTOP...



Task 4: Configure R3 to Support SSH Connections

1. Configure a domain name of **ccnasecurity.com** on R3.

```
R3(config)#ip domain-name ccnasecurity.com
```

2. Configure users for login to the SSH server on R3. Create a user ID of **SSHadmin** with the highest possible privilege level and a secret password of **ciscosshpa55**.

```
R3(config)# username SSHadmin privilege 15 secret ciscosshpa55
```

3. Configure the incoming vty lines on R3. Use the local user accounts for mandatory login and validation. Accept only SSH connections.

```
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#transport input ssh
R3(config-line)#exit
```

4. Erase existing key pairs on R3. Any existing RSA key pairs should be erased on the router.

```
R3(config)# crypto key zeroize rsa
```

Note: If no keys exist, you might receive this message: % No Signature RSA Keys found in configuration.

```
R3(config)#crypto key zeroize rsa
% No Signature RSA Keys found in configuration.
```

5. Generate the RSA encryption key pair for R3. The router uses the RSA key pair for authentication and encryption of transmitted SSH data. Configure the RSA keys with a modulus of **1024**. The default is 512, and the range is from 360 to 2048.

```
R3(config)# crypto key generate rsa
```

```
R3(config)# crypto key generate rsa
The name for the keys will be: R3.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 2048 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
```

```
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
```



```
R3(config)#crypto key generate rsa
The name for the keys will be: R3.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.
```

```
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
```

6. Verify the SSH configuration. Use the **show ip ssh** command to see the current settings. Verify that the authentication timeout and retries are at their default values of 120 and 3.
7. Configure SSH timeouts and authentication parameters. The default SSH timeouts and authentication parameters can be altered to be more restrictive. Set the timeout to **90** seconds, the number of authentication retries to **2**, and the version to **2**. Issue the **show ip ssh** command again to confirm that the values have been changed.

```
R3#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 90 secs; Authentication retries: 2
```

8. Attempt to connect to R3 via Telnet from PC-C. Open the Desktop of **PC-C**. Select the Command Prompt icon. From **PC-C**, enter the command to connect to **R3** via Telnet.

```
PC> telnet 192.168.3.1
```

This connection should fail because **R3** has been configured to accept only SSH connections on the virtual terminal lines.

9. Connect to R3 using SSH on PC-C. Open the Desktop of **PC-C**. Select the Command Prompt icon. From PC-C, enter the command to connect to R3 via SSH. When prompted for the password, enter the password configured for the administrator **ciscosshpa55**.

```
PC> ssh -l SSHadmin 192.168.3.1
```

10. Connect to R3 using SSH on R2. To troubleshoot and maintain R3, the administrator at the ISP must use SSH to access the router CLI. From the CLI of **R2**, enter the command to connect to **R3** via SSH version **2** using the **SSHadmin** user account. When prompted for the password, enter the password configured for the administrator: **ciscosshpa55**.

```
R2# ssh -v 2 -l SSHadmin 10.2.2.1
```



```
Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 192.168.3.1
Trying 192.168.3.1 ...Open

[Connection to 192.168.3.1 closed by foreign host]
C:\>ssh -l SSHadmin 192.168.3.1
Invalid Command.

C:\>ssh -l SSHadmin 192.168.3.1
Invalid Command.

C:\>ahow ip ssh
Invalid Command.

C:\>telnet 192.168.3.1
Trying 192.168.3.1 ...Open

[Connection to 192.168.3.1 closed by foreign host]
C:\>ssh -l SSHadmin 192.168.3.1

Password:
% Password: timeout expired!
% Login invalid

[Connection to 192.168.3.1 closed by foreign host]
C:\>ssh -l SSHadmin 192.168.3.1

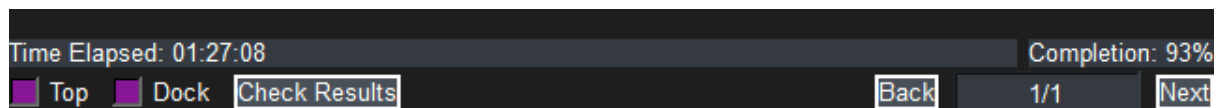
Password:

R3#ssh -v 2 -l SSHadmin 10.2.2.1

Password:

R3#
```

11. Check results. Your completion percentage should be 100%. Click Check Results to view the feedback and verification of which required components have been completed



Assessment Rubric

Lab 03

Configure Cisco Routers for Syslog, NTP and SSH Operations



Name:	Student ID:
-------	-------------

Points Distribution

Task No.	LR 2 Simulation	LR9 Report
Task 1	20	
Task 2	20	
Task 3	10	
Task 4	30	
Total	/80	/10
CLO Mapped	CLO 1	CLO1

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission	/10	CLO 1

CLO	Total Points	Points Obtained
1	100	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

**Lab Evaluation Assessment Rubric**

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3: Good Points 3	Level 4: Exemplary Points 4
LR2	Program/Code / Simulation Model/ Network Model	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR9	Report	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.